Precursor Groups and the Inhour Equation

NE 630 | Nuclear Reactor Theory

28
LESSON

Read: FNRP 5.4–5.5, especially Table 5.1.

Construct a grouped-precursor model, derive the inhour equation, and compare reactor periods from one- and six-group models.

1 From one nuclide to precursor groups

Many nuclides produce delayed neutrons. For thermal fission of U-235, the ten examples below account for about 72% of delayed neutrons in the cited evaluation. The last column gives each nuclide's percentage of the total delayed-neutron yield, not a percentage of all fission neutrons.

Precursor	$t_{1/2}$ (s)	Share of delayed yield (%)
^{137}I	24.6	16.3
^{89}Br	4.34	11.1
^{94}Rb	2.70	9.3
^{90}Br	1.91	7.4
^{88}Br	16.3	7.3
^{85}As	2.02	5.3
^{138}I	6.25	4.7
^{139}I	2.28	3.5
^{95}Rb	0.378	3.5
^{87}Br	55.6	3.2
Subtotal		**71.6**

Yield shares: P. Dimitriou et al., *Nuclear Data Sheets* 173, 144–238 (2021), Table VII, U-235 thermal, CRP column (CRP+ENDF/B-VIII.0 decay data; JEFF-3.1.1 fission yields). Half-lives rounded from Tables I–II; As-85 from B. Singh and J. Chen, ENSDF (2014).

Average by delayed-neutron yield. Let s_j be the delayed-yield share of nuclide j. Normalize the shares over the precursors being averaged:

$$\bar{t}_{1/2} = \frac{\sum_j s_j t_{1/2,j}}{\sum_j s_j}, \qquad \bar{\lambda} = \frac{\ln 2}{\bar{t}_{1/2}}.$$

This preserves the mean delay $\bar{t}_{1/2}/\ln 2$. For the ten listed nuclides, divide by 71.6: $\bar{t}_{1/2} = 11.66$ s and $\bar{\lambda} = 0.05946\ \text{s}^{-1}$ *for this subset.* The remaining 28.4% is needed for a full U-235 average.

Averaging procedure: FNRP Eqs. (5.32)–(5.34).

Represent the combined release of all precursors with G effective groups, each with delayed fraction β_i and decay constant $\lambda_i = \ln 2/t_{1/2,i}$. The values below are from FNRP Table 5.1.

		Delayed fraction β_i		
i	$t_{1/2,i}$ (s)	U-233	U-235	Pu-239
1	56	0.00023	0.00021	0.00007
2	23	0.00078	0.00142	0.00063
3	6.2	0.00064	0.00128	0.00044
4	2.3	0.00074	0.00257	0.00069
5	0.61	0.00014	0.00075	0.00018
6	0.23	0.00008	0.00027	0.00009
$\beta = \sum_i \beta_i$		0.00261	0.00650	0.00210
$\bar{\nu}$ (neutrons/fission)		2.50	2.43	2.90

Each C_i is an **effective delayed-neutron reservoir**. Unlike Lesson 27's $C = f_n N_{\text{Br}}$, a grouped C_i alone does not determine an individual nuclide's concentration.

2 Equations and initial conditions

With $\beta = \sum_{i=1}^{G} \beta_i$,

$$\boxed{\begin{aligned} \frac{dn}{dt} &= \frac{\rho(t) - \beta}{\Lambda} n + \sum_{i=1}^{G} \lambda_i C_i + S(t), \\ \frac{dC_i}{dt} &= \frac{\beta_i}{\Lambda} n - \lambda_i C_i, \qquad i = 1, \ldots, G. \end{aligned}}$$

For an initially critical, source-free equilibrium,

$$\rho(0^-) = 0, \qquad n(0) = n_0, \qquad \boxed{C_i(0) = \frac{\beta_i n_0}{\lambda_i \Lambda}.}$$

These equations have the matrix form $\dot{\mathbf{y}} = \mathbf{A}(\rho)\mathbf{y} + \mathbf{b}$, where $\mathbf{y} = [n, C_1, \ldots, C_G]^T$ and $\mathbf{b} = [S, 0, \ldots, 0]^T$. The 7×7 matrix for six groups has

$$A_{00} = (\rho - \beta)/\Lambda, \quad A_{0i} = \lambda_i, \qquad A_{i0} = \beta_i/\Lambda, \quad A_{ii} = -\lambda_i$$

and all other entries zero.

Check. With U-235, $n_0 = 10^8$ cm^{-3}, and $\Lambda = 5 \times 10^{-5}$ s, calculate $C_1(0)$. Why is setting all $C_i(0) = 0$ a different physical experiment?

$$C_1(0) = _____ \text{ cm}^{-3}.$$

Reactor periods and a numerical comparison

3 The pathway to the inhour equation

After a step, take ρ constant and $S = 0$. **Try one exponential mode:**

$$n(t) = ae^{\omega t}, \qquad C_i(t) = b_i e^{\omega t}.$$

Eliminate the precursor amplitudes:

$$(\omega + \lambda_i)b_i = \frac{\beta_i}{\Lambda}a \quad \Longrightarrow \quad b_i = \frac{\beta_i a}{\Lambda(\omega + \lambda_i)}.$$

Substitute into the neutron equation:

$$\Lambda\omega = \rho - \beta + \sum_{i=1}^{G} \frac{\beta_i \lambda_i}{\omega + \lambda_i}.$$

Use $\beta = \sum_i \beta_i$ and rearrange:

$$\boxed{\rho = \Lambda\omega + \sum_{i=1}^{G} \frac{\beta_i \omega}{\omega + \lambda_i}.}$$

This is the **inhour equation**. It is rational; clearing the denominators gives the degree-$(G + 1)$ characteristic polynomial. The poles $-\lambda_i$ are not roots.

The full solution is a sum of $G + 1$ modes, with amplitudes fixed by the initial state. The largest eigenvalue $\omega_\star$ determines the late-time signed period $T = 1/\omega_\star$:

$$\boxed{\rho = \frac{\Lambda}{T} + \sum_{i=1}^{G} \frac{\beta_i}{1 + \lambda_i T}.}$$

At criticality, $\omega_\star = 0$ and the equilibrium period is infinite. For a negative insertion, $T < 0$.

4 One effective group and useful limits

Near criticality, if $|\omega_\star| \ll \lambda_i$ for every group,

$$T \simeq \frac{\Lambda + \sum_i \beta_i/\lambda_i}{\rho}.$$

A one-group model using Section 1's average preserves the equilibrium total precursor inventory:

$$\beta_{1g} = \beta, \qquad \lambda_{1g} = \bar{\lambda} = \frac{\beta}{\sum_i \beta_i/\lambda_i}.$$

For the U-235 groups on page 1, $\bar{t}_{1/2} = 9.044$ s and $\bar{\lambda} = 0.076641\ \text{s}^{-1}$. This reproduces the small-reactivity limit. A mixture still has several modes; one effective group cannot reproduce every transient. This is distinct from L27's Br-87-only model.

Well above prompt criticality,

$$T \simeq \frac{\Lambda}{\rho - \beta}.$$

This estimate requires the delayed term to be small relative to $\rho - \beta$; the full period is finite at $\rho = \beta$.

For a sufficiently large negative insertion,

$$\omega_\star \longrightarrow -\lambda_1, \qquad T \longrightarrow -\frac{1}{\lambda_1} = -80.79\ \text{s}.$$

Earlier decay can be much faster than this late-time limit.

5 Work with `ne630_kinetics`

Save `ne630_kinetics.py` beside your notebook or script. Use Python with NumPy and SciPy; add Matplotlib for plots. Import it as `import ne630_kinetics as kin`. The defaults are the six U-235 groups, $\Lambda = 5 \times 10^{-5}$ s, critical initial equilibrium, $S = 0$, and no feedback. Reactivity arguments are dimensionless: $\rho = d$ `kin.BETA` for an insertion of d dollars. Times are in seconds.

From a terminal, run `python ne630_kinetics.py` to print period tables and checks; it also writes `numerical_results.json` beside the module. Import the module in your own notebook, script, or interactive Python session to vary the cases and make plots.

For a step, use `kin.sample_times(tmax)` and `kin.solve_step(rho, times)`. The result `sol` contains times in `sol.t`, n/n_0 in `sol.y[0]`, and C_i/n_0 in `sol.y[i]` ($i = 1, \ldots, 6$). `kin.dominant_period(rho)` returns the signed period. For one group, pass `beta=[kin.BETA]` and `lam=[lambda_eff]` to both functions, with `lambda_eff` set to the full six-group $\bar{\lambda}$ from Section 4. Each model initializes its own critical precursor inventory.

6 Compare one group with six

Verify equilibrium. Set $\rho = 0$: all densities should remain constant. Inspect `kin.matrix` to connect the implementation to Section 2.

Compute periods. Run $\rho/\beta = 0.02$ and 0.2. Compare the six-group period, the effective one-group period, and the small-reactivity estimate in Section 4.

ρ/β	T_{6g} (s)	T_{1g} (s)
0.02		
0.2		

Use `kin.inhour_residual` to check that each six-group frequency $1/T$ satisfies the inhour equation at the imposed ρ.

Plot and explain. Plot both 0.2β traces over 0–200 s on semilog axes. Compare the late slope with $1/T$. Why does preserving the mean delay and equilibrium inventory fail to preserve the whole transient? Which insertion is better represented by one effective group?

Notebook enrichment. Continue in `lesson_28_exploration.ipynb` for the closer-to-critical 0.002β case, prompt-critical crossings, negative-reactivity tails, and finite pulses that demonstrate precursor memory. The notebook uses the same module and six-group data.